

Problem Set 6

Each student must submit your own **HANDWRITTEN** work. Turn in a paper copy at the beginning of lecture on 11/13/2026. Please write your name on each page and make sure your problem set is stapled. Students are allowed to work in groups, but each student must turn in their own set of answers. No help from outside of the course is allowed. To receive full credit for a problem, all work must be shown. Partial credit will be awarded. All problems are weighted evenly.

1. Suppose that a monopolist faces the market demand curve $P(Q) = 40 - Q$ and has the faces the following total cost function $TC(Q) = 100 + \frac{3}{2}Q^2$.
 - (a) Write the total revenue, marginal revenue, and marginal cost functions for the monopolist.
 - (b) On a single graph, plot the demand, margin revenue, and marginal cost curves. Indicate on the graph where the markup above marginal cost is.
 - (c) Write and solve the monopolist's profit maximization problem for their choice of Q and P .
 - (d) What is the monopolist's markup above marginal cost

2. Suppose there are three firms in a market have agreed to operate as a cartel. Their total cost functions are given by $TC_1(Q_1) = 2Q_1^2$, $TC_2(Q_2) = 2Q_2 + Q_2^2$, and $TC_3(Q_3) = 6Q_3 + \frac{Q_3^2}{2}$ where subscripts 1, 2, and 3 indicate firm 1, firm 2, and firm 3 respectively. The market demand curve is given by $P(Q) = 64 - \frac{Q}{7}$

(a) Solve the cartel's profit maximization problem for their choice of Q and P .

- (b) How will production be divided among the firms in the cartel? Solve for Q_1 , Q_2 , and Q_3 .

3. Suppose that a homogeneous products duopoly faces a market demand curve given by $P = 140 - Q$ where $Q = Q_1 + Q_2$. Both firms have a constant marginal cost given by $MC = 20$.

(a) Write each firm's residual demand curve.

(b) Find each firm's marginal revenue curve.

(c) Find each firm's best response function to their competitor's level of output.

- (d) What is firm 1's profit-maximizing quantity, given that firm 2 produces an output of 30 units per year? What is firm 1's profit maximizing quantity when firm 2 produces 40 units per year?
- (e) What is the Cournot equilibrium choice of quantity for each firm? What is the price in this equilibrium?